2000m Mini Laser Rangefinder Module
Product manual



STA-M series miniature laser rangefinder module adopts 905nm semiconductor laser components, with low power consumption, small size, stable performance and other characteristics, suitable for thermal imaging, night vision and other handheld mobile devices and miniaturized UAV pod equipment integration.

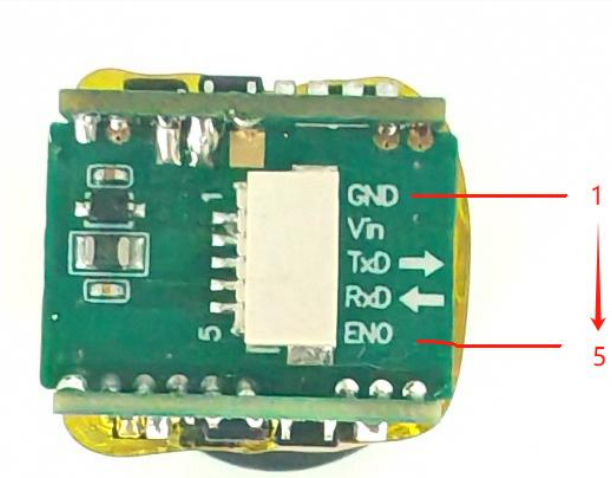
Measurement range: 5-2000meters
Measurement of trees ≥1500 meters
Measurement accuracy ±1m
Weight ≤14 ±0.5g

OEM/ODM Laser Rangefinder Module

Technical parameters are shown in the following table

Model		STA-M020X
Laser wavelength		905nm
Ranging capability	Building	2000m
	Trees	1500m
Laser grade		Class 1 Eye safety
Mini range		5m
Resolution		±0.1m
Operating Current		≤90mA
Standby current		≤20mA
Power consumption		Standby ≤0.18W
		Peak ≤0.55W
Communication Interface		TTL
Operating Temperature		-20℃~+55℃
Storage Temperature		-30℃~+60℃
Input Voltage		3-5V (Recommended 5V power supply)
Beam divergence angle		5 mard
Ranging Frequency		1-3Hz
Power		≤1 mW Safe for human eyes
Ranging method		Pulse
Weight		14±0.5g
Size		Φ23x47mm

Interfaces and pins are defined as shown in the following table

TTL interface			
Pin	Pin Definition	Describe	Schematic drawing
1	GND	Power supply -	
2	Vin	Power supply +	
3	TXD	Transmit Data	
4	RXD	Received Data	
5	ENO	Enable Pin, Reserved	

Module Communication Command Chart

Master Command		
Command code	Description	Broadcast add. response
0x80	Read rangefinder status	NO
0x81	Read measuring result	NO
0x83	Start ranging	NO
0x84	Stop ranging	NO
0x9E	Communication parameter program	YES
0xBE	Read communication parameter	YES
0xC0	Communication test	YES
0xC1	Read Firmware Version	YES
Slave Response		
Recognition code	Description	
0x60	Simple response frame	
0x01	Range data frame	

0x20	rangefinder status frame
0x21	Communication parameter frame
0xE0	Error message frame

NOTE:

1. The Communication Frame Format:

   SlaveID + instruction + parameter +checksum

2. The interval between two adjacent frames has to be at least the time of 1.5 byte when sent out, when it is less than 1ms, it is regarded as 1ms;

   600BPS: 25ms

   1200BPS: 12.5ms

   2400BPS: 6.25ms

   4800BPS: 3.125ms

   9600BPS: 1.56ms

   19200BPS: 1ms(0.78ms < 1ms, take 1ms)

3. Sending calculate of checksum;

   This module uses the verification mode of byte by byte accumulation sum, SlaveID is not included in the verification, sending checksum at last.

   Such as calculate check sum of scanning mode instruction"0x83 0x40":

   A、Calculate accumulation sum ： 0x83 + 0x40 = 0xC3；

   B、Take LSB[D7..D0]：0xC3；

   C、Calculate 2's complement：0x100 - 0xC3 = 0x3D， 0x3D is checksum；

   Add slave address when mater sends: 0x10 0x83 0x40 0x3D

4. Verification receipt;

   Such as receive communication frame: 0x10 0x83 0x40 0x3D, verification not including slave address 0x10

A、Calculate accumulation sum：0x83 + 0x40 + 0x3D = 0x100；

B、Take LSB of accumulation sum: 0x00; 0x00 is verification result；

If verification result is 0x00, that means received data is right; If verification result is not 0x00, that means received data is wrong.

Module Command Description

Note:In the example, the slave address of each command is 0x10;in application, the slave address will be the programmed one.

1、Communication test（0xC0）；

Master sends		
Command code	Parameter	Description
0xC0	None	
Slave responses		
Recognition code	Parameter	Description
0x60	None	
Example		
Master sends：0x10 0xC0 0x40；0x40 is checksum Slave response：0x10 0x60 0xA0；0xA0 is checksum		

2、Command slave to start ranging (0x83)；

Master sends		
Command code	Parameter	Description
0x83	byte1	D7: Reserve
		D6: Continuous ranging mode, 0 = off; 1 = on;
		D5: Fog mode, 0 = off; 1 = on;
		D4: Data unit, 0 = meter; 1 = yard;
		D3 - D0: Working Mode, 0 = ranging; 1 = speed; 2 = Small Target Mode;
Slave responses		
Recognition code	Parameter	Description

0x60	None	If slave is set as automatic export mode, slave responds 0x60 first after reception of start ranging command, after ranging is finished, it will then send out ranging result;
Example		
Master sends： 0x10 0x83 0x00 0x7D：Mode 0 is for single measure, data unit is meter; 0x10 0x83 0x40 0x3D：Mode 0 is for continuous measure, data unit is meter; 0x10 0x83 0x10 0x6D：Mode 0 is for single measure, data unit is yard; 0x10 0x83 0x20 0x2D：Mode 0 is for continuous measure, data unit is yard; Slave response: 0x10 0x60 0xA0；		

Note: 1、If module data export method is set as **query mode**, then it can only do single measure;
2、In speed mode, continuous ranging function is null, that is no continuous speed measure;
3、Fog mode is only valid for device with fog mode function;
4、Slave will send 0x60 response frame after reception of start ranging command, it will start to send ranging result after ranging is finished.

3、Command slave stop ranging (0x84)；

Master sends		
Command code	Parameter	Description
0x84	None	

Note: After master sends command to slave to do continuous ranging, it needs to send this command to slave to stop ranging. If master sends command to slave to do single time ranging, then no need to send this command, cause slave will stop ranging automatically after ranging is finished.

4、Read slave status (0x80)；

Master sends		
Command code	Parameter	Description
0x80	None	
Slave responses		
Recognition code	Parameter	Description
0x20	byte1	D7：1 = Rangefinder Busy; 0 = Ranging finished; D6: 1=Rangefinder Malfunction; 0=No malfunction; D5-D2: Reserve; D1: 0=Angle sensor prohibited; 1=Angle sensor enabled; D0: 0=Angle sensor is normal; 1=Abnormal Angle sensor;

	byte2	Reserve

5、Read range finder result (0x81)；

Master sends		
Command code	Parameter	Description
0x81	None	
Slave responses		
Recognition code	Parameter	Description
0x01	byte1	D7:： Distance data, 0=data valid; 1=Invalid data; D6:： Angle data, 0=data valid; 1=Invalid data; D5： Data resolution, 0 = 0.5/LSB； 1 = 0.1/LSB; D4： Data unit, 0 = meter; 1 = yard; D3 - D0： Working Mode, 0 = Ranging; 1 = Speed;
	byte2	When distance is valid： Distance[D15..D8]； When distance is invalid： 0x80 = Ranging no result; 0x81= System error;
	byte3	When angle is valid: Distance[D7..D0]； When angle is invalid: Reserve;

Example

1、slave sends distance frame: 0x10 0x01 0x60 0x12 0xD7 0xB6
distance value = 0x12D7 = 4823 = 482.3m

2、slave sends speed frame： 0x10 0x01 0x21 0x03 0xD7 0x04
speed value = 0x03D7 = 983 = 98.3km/h

3、slave sends distance frame： 0x10 0x01 0xE0 0x80 0x00 0x9F
The distance value is invalid (the first byte is 0xE0, D7=1)

6、Write slave communication baud rate (0x9E),The factory setting is 19200BPS；

Master sends		
Command code	Parameter	Description
0x9E	byte1	has to be 0x90
	byte2	Reserve

	byte3	Reserve
	byte4	baud rate setting： 0 = 1200BPS; 1 = 2400BPS; 2 = 4800BPS; 3 = 9600BPS; 4 = 19200BPS(Default)；
Slave responses		
Recognition code	Parameter	Description
（please refer to the slave response part of reading slave parameter）		
Example		
If master sends frame: 0x10 0x9E 0x90 0x00 0x00 0x04 0xCE; It's means: Set baud rate of slave as 19200BPS,		

Note:New baud rate is only valid after module is restarted;

7、Write slave address (0x9E), Factory settings: 0x10；

Master sends		
Command code	Parameter	Description
0x9E	byte1	Must be 0x91
	byte2	Reserve
	byte3	Reserve
	byte4	slave new address;
Slave responses		
Recognition code	Parameter	Description
（please refer to the slave response part of reading slave parameter）		
Example		
To write slave address as 0x20, then master sends: 0x10 0x9E 0x91 0x00 0x00 0x20 0xB1		

Note: 1. New slave address is valid after the device is restarted
2. 0x00 is broadcast address, so slave address can't be set as 0x00;
3. Default slave address is 0x10;

8、Write measured result export method (0x9E)；

Master sends		
Command code	Parameter	Description

0x9E	byte1	Must be 0x92
	byte2	Reserve
	byte3	Reserve
	byte4	Measured result export method selection: 0 = **Query mode**(slave can only export data when master sends reading command) 1(Default) = Automatic mode (After ranging finished, it sends out data automatically);
Slave responses		
Recognition code	Parameter	Description
（Please refer to the slave response part of reading slave parameter）		
Example		
To change data export to automatic exporting, then master sends： 0x10 0x9E 0x92 0x00 0x00 0x01 0xCF		

9、Set UART IO voltage (0x9E),Factory setting: 3.3V；

Example		
Command code	Parameter	Description
0x9E	byte1	Must be 0x93
	byte2	Reserve
	byte3	Reserve
	byte4	UART IO voltage Setting： 72 - 199， 0.025V/LSB；
Slave responses		
Recognition code	Parameter	Description
（please refer to the slave response part of reading slave parameter）		
Example		
Change UART IO voltage to 3.3V， then master sends： 0x10 0x9E 0x93 0x00 0x00 0x84 0x4B		

Note： VIO = Setting * 0.025V, Default VIO = 3.3V

10、Reading communication parameter (0xBE)；

Master sends		
Command code	Parameter	Description

0xBE	byte1	Must be 0x91 or 0x92
Slave responses		
Recognition code	Parameter	Description
0x21	byte1	**Slave address**: 0x01 - 0xFF
	byte2	D4: Data export method ; D3 - D0： Baud Rate setting；
	byte3	UART IO voltage setting

11、Slave sends wrong information（0xE0）；

Slave responses		
Recognition code	Parameter	
0xE0	byte1	D7 - D3: Reserve； D2： 1 = Communication parameter error； D1： 1 = Unrecognizable command； D0： 1 = Checksum error；

12、Read Firmware Version；

Master Sends		
Command Code	Parameter	
0xC1	None	
Slave responses		
Recognition code	Parameter	Notes
0x61	byte0	Major Version (BCD)
	byte1	Minor Version (BCD)
Example		
Slave responses: 0x10 0x61 0x07 0x00 0x98 Data "0x07 0x00" means firmware version is V7.00.		